



UNIVERSITÉ PARIS-SACLAY

Rough and Lifted Bergomi Models for SPX and VIX Option Pricing

Master 2 : Quantitative Finance

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1 Introduction

The modeling of financial markets has undergone significant developments over the past decades, driven by the need to accurately capture empirical features observed in option prices. A natural starting point is the classical Black-Scholes model, which assumes that the underlying asset follows a geometric Brownian motion with constant volatility. When inverting the Black-Scholes formula using market option prices, one obtains an implied volatility surface that depends on both strike and maturity, exhibiting characteristic patterns such as smiles and skews.

This empirical observation motivated the development of more sophisticated models such as stochastic volatility models, where volatility itself evolves as a stochastic process. Among these, the Heston model stands out as a seminal contribution, where the variance follows a mean-reverting square-root process, inspired by earlier Vasicek-type dynamics. Due to its analytical tractability, in particular the availability of a closed-form characteristic function, the Heston model became a benchmark in the literature.

However, empirical studies revealed that volatility exhibits a much rougher behavior than what is implied by classical diffusion-based models. In particular, volatility trajectories display a fractional regularity, typically characterized by a Hurst parameter significantly smaller than $1/2$. This observation led to the emergence of rough volatility models, where volatility is driven by fractional processes, providing a better fit to both historical data and implied volatility surfaces (see, e.g., [1]).

Nevertheless, rough models are inherently non-Markovian and pose significant computational challenges. To overcome these difficulties, lifted (or multi-factor) approximations have been introduced, which approximate rough dynamics through a finite number of mean-reverting factors, thus recovering a Markovian structure while preserving key rough features (see, e.g., [3]).

In this project, we focus on the rough Bergomi framework and its lifted counterparts, which provide a tractable and accurate alternative to classical models. We begin by briefly introducing the theoretical foundations and we then move to their numerical implementation, detailing Monte Carlo simulation procedures for option pricing. The models are subsequently calibrated to real market data for SPX and VIX options, with the objective of reproducing the observed implied volatility smiles and assessing their performance in practical pricing applications.

2 Bergomi models

Let $(S_t)_{t \geq 0}$ denote the asset price process under the risk-neutral measure. Its dynamics are given by

$$dS_t = S_t \left(r dt + \sqrt{V_t} dB_t \right) \quad (1)$$

where V_t is the instantaneous variance process and B_t is a Brownian motion. A key feature of the Bergomi model is that the variance process is constructed from the forward variance curve. Define the forward variance

$$\xi_t(u) = \mathbb{E}[V_u \mid \mathcal{F}_t], \quad u \geq t \quad (2)$$

with initial condition $\xi_0(u)$. In the Bergomi model with N factors, the spot variance is given by

$$V_t = \xi_0(t) \exp \left(\sum_{i=1}^N c_i X_t^{(i)} + \text{drift} \right) \quad (3)$$

where the factors $X_t^{(i)}$ are Ornstein-Uhlenbeck processes of the form

$$X_t^{(i)} = \int_0^t e^{-x_i(t-s)} dW_s^{(i)} \quad (4)$$

or equivalently

$$dX_t^{(i)} = -x_i X_t^{(i)} dt + dW_t^{(i)} \quad (5)$$

with $x_i > 0$ the mean-reversion speeds. The Brownian motions $W^{(i)}$ may be correlated among themselves and with the Brownian motion B driving the asset price, allowing the model to capture the leverage effect observed in equity markets. The forward variance process can also be written in exponential form as

$$\xi_t(u) = \xi_0(u) \exp \left(\sum_{i=1}^N c_i \int_0^t e^{-x_i(u-s)} dW_s^{(i)} + \text{drift} \right) \quad (6)$$

which highlights that the model is constructed as a lognormal process driven by a weighted sum of Ornstein-Uhlenbeck factors.

In addition, there is a link between the forward variance curve and observable market quantities. In particular, the fair strike $VS(t, T)$ of a variance swap satisfies

$$(T - t)VS(t, T) = \int_t^T \mathbb{E}^Q[V_s | \mathcal{F}_t] ds \quad (7)$$

which implies

$$\frac{\partial}{\partial T} ((T - t)VS(t, T)) = \mathbb{E}^Q[V_T | \mathcal{F}_t] = \xi_t(T) \quad (8)$$

This provides a practical way to infer the initial forward variance curve $\xi_0(T)$ from market data, either through variance swaps or via option-based replication techniques.

Moreover, the model yields an explicit approximation for the term structure of the at-the-money implied volatility skew. Defining

$$\psi(\tau) = \left. \frac{\partial}{\partial k} \sigma_{BS}(k, \tau) \right|_{k=0} \quad (9)$$

where $\tau = T - t$ denotes the time to maturity and $k = \log \left(\frac{K}{S_t} \right)$ is the log-moneyness, one obtains

$$\psi(\tau) \sim \sum_{i=1}^N \frac{c_i \rho_i}{2x_i \tau} \left(1 - \frac{1 - e^{-x_i \tau}}{x_i \tau} \right) \quad (10)$$

where ρ_i denotes the correlation between the factor $W^{(i)}$ and the Brownian motion driving the asset price. This expression highlights that the term structure of the skew is governed by the exponential kernels used in the factor representation.

2.1 Rough Bergomi model

Empirical studies have shown that classical stochastic volatility models fail to reproduce the observed behavior of the at-the-money (ATM) implied volatility skew, in particular its power-law decay as a function of maturity. In the Bergomi model, the skew is driven by exponential kernels, leading to a decay that is too smooth compared to market data. This motivates the introduction of rough volatility models, where exponential kernels are replaced by power-law kernels.

2.1.1 Model dynamics

The variance process is defined as

$$V_t = \xi_0(t) \exp \left(\eta \sqrt{2H} \int_0^t (t-s)^{H-\frac{1}{2}} dW_s - \frac{1}{2} \eta^2 t^{2H} \right) \quad (11)$$

where $\eta > 0$ is the volatility of volatility parameter. This can be written in terms of a fractional Brownian motion (fBM) as

$$V_t = \xi_0(t) \exp \left(2\nu C_H \widetilde{W}_t^H + \text{drift} \right) \quad (12)$$

where

$$\widetilde{W}_t^H = \int_0^t (t-s)^{H-\frac{1}{2}} dW_s \quad (13)$$

is a Volterra-type process closely related to fractional Brownian motion.

2.1.2 Fractional Brownian motion

Introduced by Kolmogorov, it is a centered Gaussian process $(W_t^H)_{t \geq 0}$ with covariance function

$$\mathbb{E}[W_t^H W_s^H] = \frac{1}{2} (t^{2H} + s^{2H} - |t-s|^{2H}) \quad (14)$$

where $H \in (0,1)$ is the Hurst parameter. This parameter governs both the regularity of the trajectories and the dependence structure of the increments: for $H = 1/2$ one recovers standard Brownian motion, while $H < 1/2$ corresponds to negatively correlated (rough) increments and $H > 1/2$ to positively correlated increments. A fundamental representation of fBM is the Mandelbrot-Van Ness formula, given by

$$W_t^H = C_H \left(\int_0^t (t-s)^{H-\frac{1}{2}} dW_s + \int_{-\infty}^0 \left[(t-s)^{H-\frac{1}{2}} - (-s)^{H-\frac{1}{2}} \right] dW_s \right) \quad (15)$$

where C_H is a normalization constant. This representation highlights that fBM is a Gaussian process with long memory, built from a fractional kernel acting on a Brownian motion. In practice, rough volatility models rely on a simplified version of this, known as the Riemann-Liouville fractional Brownian motion,

$$W_t^{H, \text{RL}} = \frac{1}{\Gamma(H + \frac{1}{2})} \int_0^t (t-s)^{H-\frac{1}{2}} dW_s \quad (16)$$

which corresponds to retaining only the causal part of the Mandelbrot-Van Ness decomposition. This process is a particular example of a Volterra Gaussian process, i.e., a process of the form

$$X_t = \int_0^t K(t,s) dW_s \quad (17)$$

with kernel $K(t,s) = (t-s)^{H-\frac{1}{2}}$. While $W_t^{H, \text{RL}}$ shares the same local regularity as fractional Brownian motion, it does not have stationary increments and therefore is not strictly an fBM. However, it preserves the key feature of roughness for $H < 1/2$ and, importantly, it is adapted and causal, making it suitable for financial modeling.

2.1.3 Non-Markovian and non-affine structure

Indeed, the rBergomi model is not Markovian in the spot variance, in the sense that

$$\mathbb{E}[V_u | \mathcal{F}_t] \neq \mathbb{E}[V_u | V_t] \quad (18)$$

However, it can be viewed as Markovian in the infinite-dimensional state variable given by the forward variance curve $\xi_t(u)$.

Moreover, unlike the Heston model, the rough Bergomi framework is not affine, which precludes the use of Fourier-based pricing techniques. Therefore, option pricing must be performed using Monte Carlo methods.

2.1.4 Interpretation of the Hurst parameter

A key feature of the model is that the Hurst parameter H has a direct financial interpretation. Indeed, it controls the decay of the ATM implied volatility skew for short maturities:

$$\psi(\tau) \sim \tau^{H-\frac{1}{2}} \quad (19)$$

This provides a clear link between the roughness of volatility and observable market quantities and it was used in the code to obtain an initial estimate of the value of H prior to calibration.

2.2 Lifted Bergomi model

The Lifted Bergomi model circumvents the difficulties of the rBergomi by replacing the singular fractional kernel with a finite sum of exponentials, thereby lifting the process into a finite-dimensional Markovian system of SDEs.

2.2.1 Laplace representation and factor decomposition

The key observation underlying the Lifted Bergomi approximation is that the fractional kernel $K(t) = t^{\alpha-1}/\Gamma(\alpha)$ admits a Laplace representation. By the Bernstein-Widder theorem, a completely monotone kernel on $(0, \infty)$ can be written as the Laplace transform of a positive measure μ :

$$K(t) = \int_0^\infty e^{-xt} \mu(dx), \quad t > 0 \quad (20)$$

For the fractional kernel with $\alpha = H + \frac{1}{2} \in (\frac{1}{2}, 1)$, this measure is absolutely continuous with density

$$\mu(dx) = \frac{x^{-\alpha}}{\Gamma(\alpha) \Gamma(1-\alpha)} dx \quad (21)$$

Substituting (20) into the definition of $\widetilde{W}$ and applying the stochastic Fubini theorem yields

$$\widetilde{W}_t = \int_0^\infty U_t(x) \mu(dx) \quad (22)$$

where, for each $x \geq 0$, the factor process $U_t(x)$ is defined by

$$U_t(x) = \int_0^t e^{-x(t-s)} dW_s \quad (23)$$

Crucially, for each fixed x , $\{U_t(x)\}_{t \geq 0}$ is a mean-reverting Ornstein-Uhlenbeck process satisfying the SDE

$$dU_t(x) = -x U_t(x) dt + dW_t, \quad U_0(x) = 0 \quad (24)$$

which is a semimartingale even though $\widetilde{W}$ itself is not.

2.2.2 Finite-dimensional approximation

We approximate the measure μ by a discrete measure supported on n points:

$$\mu(dx) \approx \mu_n(dx) = \sum_{i=1}^n c_i \delta_{x_i}(dx) \quad (25)$$

so that the kernel is approximated by a sum of exponentials:

$$K(t) \approx K_n(t) = \sum_{i=1}^n c_i e^{-x_i t} \quad (26)$$

Correspondingly, the Volterra process $\widetilde{W}$ is approximated by the finite sum

$$Y_t = \sum_{i=1}^n c_i U_t^i \quad (27)$$

where each factor $U_t^i \equiv U_t(x_i)$ satisfies

$$dU_t^i = -x_i U_t^i dt + dW_t, \quad U_0^i = 0, \quad i = 1, \dots, n \quad (28)$$

The system $(U_t^1, \dots, U_t^n)$ is an n -dimensional Markov process driven by a single Brownian motion W , and the resulting model is a classical (multi-factor) Bergomi model with a specific parametrization of the weights and mean-reversion speeds.

2.2.3 Geometric partition for the fractional kernel

To choose the nodes x_i and weights c_i in (25), we partition the positive real line using an increasing sequence $0 = \eta_0 < \eta_1 < \dots < \eta_n$ and define

$$c_i = \int_{\eta_{i-1}}^{\eta_i} \mu(dx), \quad x_i = \frac{1}{c_i} \int_{\eta_{i-1}}^{\eta_i} x \mu(dx), \quad i = 1, \dots, n \quad (29)$$

That is, c_i is the μ -mass of the interval $[\eta_{i-1}, \eta_i]$ and x_i is the corresponding μ -center of mass. For the fractional kernel, a geometric partition with ratio $r > 1$ is particularly effective. Setting $\eta_i = r^{i-n/2}$ for $i = 0, 1, \dots, n$ (with n even), the closed-form expressions for the weights and nodes become

$$c_i = \frac{r^{1-\alpha} - 1}{\Gamma(\alpha) \Gamma(2-\alpha)} r^{(1-\alpha)(i-1-n/2)}, \quad i = 1, \dots, n \quad (30)$$

$$x_i = \frac{1-\alpha}{2-\alpha} \frac{r^{2-\alpha} - 1}{r^{1-\alpha} - 1} r^{i-1-n/2}, \quad i = 1, \dots, n \quad (31)$$

The centering by $n/2$ in the exponent ensures that the nodes are distributed symmetrically (on a log scale) around $r^0 = 1$, spanning both fast-reverting and slow-reverting time scales.

2.2.4 Model dynamics

Combining the finite-factor approximation with the rBergomi variance specification, the Lifted Bergomi model is defined by the following system:

$$\frac{dS_t}{S_t} = \sqrt{V_t} dB_t \quad (32)$$

$$V_t = \xi_0(t) \exp\left(a Y_t - \frac{1}{2} a^2 \text{Var}(Y_t)\right) \quad (33)$$

$$Y_t = \sum_{i=1}^n c_i U_t^i \quad (34)$$

$$dU_t^i = -x_i U_t^i dt + dW_t, \quad U_0^i = 0, \quad i = 1, \dots, n \quad (35)$$

where $d\langle B, W \rangle_t = \rho dt$, $a = \eta\sqrt{2H}$, and the variance of the aggregated process is

$$\text{Var}(Y_t) = \sum_{i=1}^n \sum_{j=1}^n c_i c_j \frac{1 - e^{-(x_i+x_j)t}}{x_i + x_j} \quad (36)$$

The model parameters are:

- $\xi_0(\cdot)$: the forward variance curve, calibrated from market ATM implied volatilities;
- $H \in (0, \frac{1}{2})$: the Hurst (roughness) parameter;
- $\eta > 0$: the volatility-of-volatility;
- $\rho \in (-1, 0)$: the spot-vol correlation;
- $n \in \mathbb{N}$ and $r > 1$: the number of factors and geometric ratio, which control the approximation quality.

For fixed n and r , the weights (c_i) and mean-reversion speeds (x_i) are deterministic functions of H through (30)-(31), so the freely calibrated parameters reduce to (H, η, ρ) , the same as in the rBergomi model.

2.2.5 Properties and relationship to rBergomi

The Lifted Bergomi model enjoys several attractive features:

- **Markovianity.** The state vector $(\log S_t, U_t^1, \dots, U_t^n)$ is a finite-dimensional Markov process, in contrast to the rBergomi model which requires the entire path history. This enables the use of standard numerical methods, like efficient Euler-type simulation schemes.
- **Convergence to rBergomi.** Under the conditions $r_n \downarrow 1$ and $n \ln(r_n) \rightarrow \infty$, the Lifted Bergomi variance process converges in law to the rBergomi variance as $n \rightarrow \infty$. In practice, moderate values such as $n = 20$ and $r = 2$ already provide an excellent approximation.
- **Rough regularity.** Although each factor U^i is a standard (Markovian) Ornstein-Uhlenbeck process with $H = \frac{1}{2}$ regularity, the weighted sum $Y_t = \sum_i c_i U_t^i$ inherits approximate H -Hölder regularity from the fractional kernel approximation. In particular, the sample paths of $\log V$ in the Lifted Bergomi model exhibit the same rough, rapidly fluctuating behavior as those of the rBergomi model, which is key to reproducing the observed power-law decay of the ATM skew.

3 Numerical implementation

After introducing the models from a theoretical perspective, this section focuses on their numerical implementation. We first describe the Monte Carlo methodology used for option pricing within these models, as well as the market data employed and the calibration procedure. Finally, we compare the results obtained and assess roughness.

3.1 Market data

All the available data were collected from the Chicago Board Options Exchange (CBOE) website as of March 16, 2026. Only the risk-free rate r , which is assumed to be constant in our modeling framework, was obtained from Yahoo Finance, using the 3-month U.S. Treasury bill as a proxy, whose value on March 16 was approximately 0.03605.

3.1.1 SPX implied volatility surface

Among the data that will certainly be useful for the calibration, there are the implied volatilities derived from SPX option prices. The website directly provided the implied volatilities, thus sparing us the inversion from market prices (a technique that will be explained in the next section). The spot value of the underlying index as of March 16 was 6706.22, which we take as our S_0 . The following option maturities were considered, all in 2026: March 23, April 16, May 29, June 30, July 31, August 31, September 30, and December 31. Each time to maturity was then annualized.

For each maturity, all available strikes were considered, although in the calibration step we will later select a subset of them. Moreover, since out-of-the-money options are generally more liquid than in-the-money ones (which tend to be held rather than traded), for strikes below the spot we considered put implied volatilities, while for strikes above the spot we considered call implied volatilities, even though the differences were minimal. In the figure below, we can observe the volatility smile, which provides an insight into the shape of the implied volatility surface.

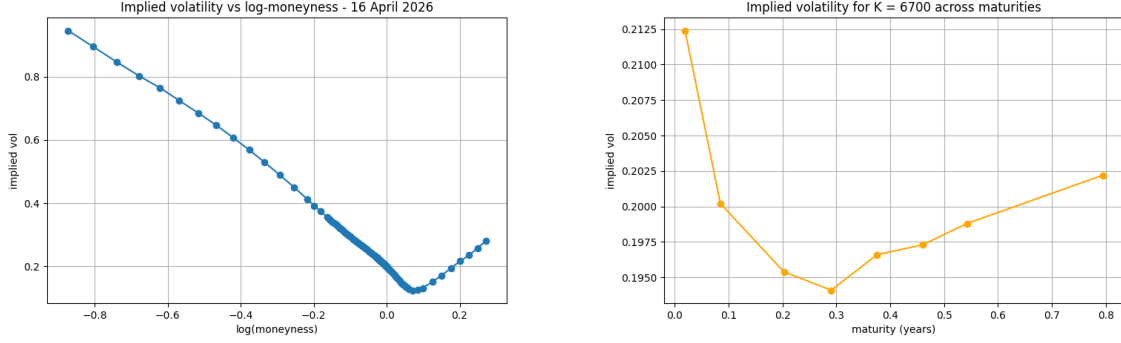


Figure 1: Sections of the volatility surface of the SPX for a fixed T (left) and for a fixed K (right)

3.1.2 VIX implied volatility surface

For VIX options, the procedure was similar to the one described above. However, we observed that fewer options are traded compared to those on the SPX. The maturities considered are: April 15, May 19, June 17, July 22, August 19, September 16, October 21, and November 18.

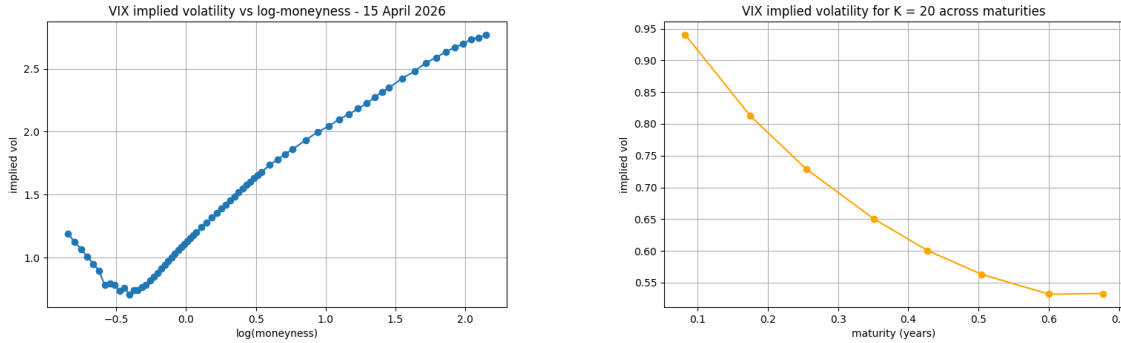


Figure 2: Sections of the volatility surface of the VIX for a fixed T (left) and for a fixed K (right)

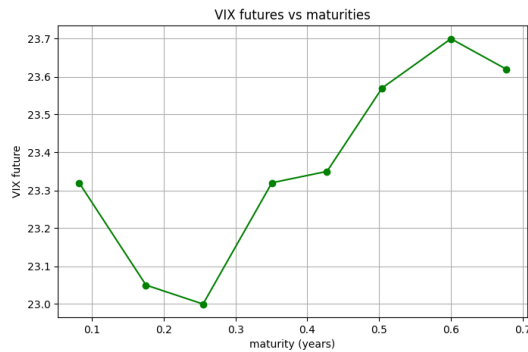


Figure 3: Value of VIX futures for the considered maturities

Moreover, the spot is not relevant in this case, since the underlying consists of VIX futures corresponding to the option maturities. For this reason, the futures prices for the same maturities were also collected and are shown in the figure above.

3.2 Pricing engines

Let's start by describing in detail the pricing method in order to make the algorithmic implementation in Python clear.

3.2.1 Forward variance curve

The forward variance curve $\xi_0(t) = \mathbb{E}[V_t]$ is extracted from market data as follows. For each available market maturity T_j , $j = 1, \dots, J$, we identify the ATM implied volatility σ_j . We then compute the total implied variance at each maturity:

$$v(T_j) = \sigma_j^2 T_j, \quad j = 1, \dots, J \quad (37)$$

anchored at $v(0) = 0$. The forward variance curve is therefore defined as the piecewise-constant derivative of the total variance (see Section 2):

$$\xi_0(t) = \frac{v(T_{j+1}) - v(T_j)}{T_{j+1} - T_j}, \quad t \in [T_j, T_{j+1}) \quad (38)$$

with the convention $T_0 = 0$ and $v(T_0) = 0$, and flat extrapolation beyond the last maturity.

3.2.2 Correlated Brownian increments

Both models require two correlated Brownian motions B (driving the price) and W (driving the variance), with $d\langle B, W \rangle_t = \rho dt$. On a uniform time grid with step Δt , we generate two independent standard normal matrices $Z_1, Z_2 \in \mathbb{R}^{N_{\text{paths}} \times N_{\text{steps}}}$ and set

$$\Delta W_k = Z_{1,k} \sqrt{\Delta t}, \quad \Delta B_k = \rho \Delta W_k + \sqrt{1 - \rho^2} Z_{2,k} \sqrt{\Delta t} \quad (39)$$

This is equivalent to a Cholesky decomposition of the 2×2 correlation matrix $\begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}$.

3.2.3 Simulation of the rBergomi model

Discretization of the Volterra process. The Volterra Gaussian process $\widetilde{W}_t = \int_0^t K(t-s) dW_s$ with fractional kernel $K(t) = t^{H-1/2}$ is discretized on a uniform grid $t_k = k \Delta t$ as

$$\widetilde{W}_{t_k} = \sum_{j=0}^{k-1} K(t_k - t_j) \Delta W_j = \sum_{j=0}^{k-1} w_{k-j} \Delta W_j, \quad w_\ell = (\ell \Delta t)^{H-1/2} \quad (40)$$

For each k , this is a weighted sum of past Brownian increments, and computing it for all $k = 1, \dots, N_{\text{steps}}$ amounts to a discrete convolution of the weight vector $(w_1, \dots, w_{N_{\text{steps}}})$ with the increment vector $(\Delta W_0, \dots, \Delta W_{N_{\text{steps}}-1})$. A naïve implementation would require $O(N_{\text{steps}}^2)$ operations per path. Instead, we exploit the convolution theorem and compute the operation via the Fast Fourier Transform (FFT). Specifically, the linear convolution is evaluated as

$$\widetilde{W}_{1:N} = \text{IFFT}\left(\text{FFT}(w) \cdot \text{FFT}(\Delta W)\right)\Big|_{1:N} \quad (41)$$

where both signals are zero-padded to length $n_{\text{fft}} \geq 2 N_{\text{steps}}$ (chosen as the smallest power of 2 satisfying this bound) to ensure that the circular convolution computed by the FFT coincides with the desired linear convolution. This reduces the complexity to $O(N_{\text{paths}} \cdot N_{\text{steps}} \log N_{\text{steps}})$.

Variance process. The instantaneous variance is

$$V_{t_k} = \xi_0(t_k) \exp\left(a \widetilde{W}_{t_k} - \frac{1}{2} a^2 \widehat{\text{Var}}(\widetilde{W}_{t_k})\right), \quad a = \eta\sqrt{2H} \quad (42)$$

where the drift correction uses the discrete variance of the convolution (rather than the continuous expression t^{2H}), for consistency with the discretization:

$$\widehat{\text{Var}}(\widetilde{W}_{t_k}) = \Delta t \sum_{\ell=1}^k w_\ell^2 = \Delta t \sum_{\ell=1}^k (\ell \Delta t)^{2H-1} \quad (43)$$

Log-price via Euler-Maruyama. The discounted log-price is simulated with the Euler-Maruyama scheme:

$$\log S_{t_{k+1}} = \log S_{t_k} + \left(r - \frac{1}{2} V_{t_k}^+\right) \Delta t + \sqrt{V_{t_k}^+} \Delta B_k \quad (44)$$

where $V_{t_k}^+ = \max(V_{t_k}, 0)$ is a positivity floor applied to avoid taking the square root of a negative number.

3.2.4 Simulation of the lifted Bergomi model

Kernel approximation parameters. As described in Section 2.2.3, the fractional kernel is approximated by a sum of n exponentials with weights $(c_i)_{i=1}^n$ and mean-reversion speeds $(x_i)_{i=1}^n$ determined by H through equations (30)-(31). In our implementation we fix $n = 20$ factors and geometric ratio $r = 2$, so that the nodes $x_i = \text{const} \cdot 2^{i-1-10}$ span the range from approximately 10^{-3} to 500, covering a wide spectrum of mean-reversion time scales. As anticipated, these parameters are not free: for each value of H proposed by the optimizer, the function `kernel_params` recomputes c_i and x_i deterministically.

Implicit-explicit Euler scheme for the factors. Each factor U^i satisfies the OU SDE. We discretize it using an implicit treatment of the mean-reversion term and an explicit treatment of the noise:

$$U_{t_{k+1}}^i = \frac{U_{t_k}^i + \Delta W_k}{1 + x_i \Delta t}, \quad U_0^i = 0, \quad i = 1, \dots, n \quad (45)$$

The implicit treatment ensures stability even for the fast-reverting factors (large x_i), which would cause instability in a standard explicit Euler scheme. Defining the decay factor $d_i = 1/(1 + x_i \Delta t)$, the update (45) can be written vectorially as

$$\mathbf{U}_{k+1} = (\mathbf{U}_k + \Delta W_k \mathbf{1}) \odot \mathbf{d} \quad (46)$$

where $\mathbf{U}_k \in \mathbb{R}^{N_{\text{paths}} \times n}$, $\mathbf{d} = (d_1, \dots, d_n)$, and $\odot$ denotes element-wise multiplication with broadcasting.

Aggregation and discrete variance correction. The approximate Volterra process is $Y_{t_k} = \sum_{i=1}^n c_i U_{t_k}^i$. By unrolling the recursion (45), each factor can be expressed as

$$U_{t_k}^i = \sum_{j=0}^{k-1} d_i^{k-j} \Delta W_j \quad (47)$$

so that

$$Y_{t_k} = \sum_{j=0}^{k-1} \alpha_{k-j} \Delta W_j, \quad \alpha_\ell = \sum_{i=1}^n c_i d_i^\ell \quad (48)$$

Since the ΔW_j are i.i.d. $N(0, \Delta t)$, the variance of Y_{t_k} in the discretization is

$$\widehat{\text{Var}}(Y_{t_k}) = \Delta t \sum_{\ell=1}^k \alpha_\ell^2 \quad (49)$$

This is precomputed once before the simulation loop: for each lag $\ell = 1, \dots, N_{\text{steps}}$, the coefficient $\alpha_\ell = \sum_i c_i d_i^\ell$ is evaluated, and the cumulative sum $\sum_{\ell=1}^k \alpha_\ell^2$ is stored. As in the rBergomi case, using the discrete variance (49) rather than the continuous formula (36) ensures that $\mathbb{E}[V_{t_k}] = \xi_0(t_k)$ holds exactly at the discrete level.

Variance and log-price. The instantaneous variance is

$$V_{t_{k+1}} = \xi_0(t_{k+1}) \exp\left(a Y_{t_{k+1}} - \frac{1}{2} a^2 \widehat{\text{Var}}(Y_{t_{k+1}})\right) \quad (50)$$

and the log-price follows the same Euler-Maruyama scheme (44) as in the rBergomi model.

3.2.5 SPX option pricing

European call option prices on the SPX are estimated by the Monte Carlo estimator. Given N_{paths} simulated terminal log-prices $\log S_T^{(m)}$ at the maturity T closest to the desired expiry on the time grid, the call price for strike K is

$$\hat{C}(K, T) = e^{-rT} \frac{1}{N_{\text{paths}}} \sum_{m=1}^{N_{\text{paths}}} \max(e^{\log S_T^{(m)}} - K, 0) \quad (51)$$

This is a direct application of the strong law of large numbers: $\hat{C}(K, T) \xrightarrow{\text{a.s.}} C(K, T)$ as $N_{\text{paths}} \rightarrow \infty$.

3.2.6 VIX computation

The VIX index at time T is defined as

$$\text{VIX}_T = 100 \sqrt{\frac{1}{\Delta} \mathbb{E} \left[\int_T^{T+\Delta} V_s \, ds \mid \mathcal{F}_T \right]}, \quad \Delta = \frac{30}{365} \quad (52)$$

Since V_s is log-normal in both models, the conditional expectation $\mathbb{E}[V_s \mid \mathcal{F}_T]$ admits a closed-form expression, so the VIX can be computed semi-analytically for each Monte Carlo path without requiring a nested simulation.

Conditional expectation in the Lifted Bergomi model. The variance at time $s > T$ is $V_s = \xi_0(s) \exp(a Y_s - \frac{1}{2} a^2 \text{Var}(Y_s))$. We decompose Y_s into a component that is measurable with respect to $\mathcal{F}_T$ and an independent future component:

$$Y_s = \underbrace{\sum_{i=1}^n c_i e^{-x_i(s-T)} U_T^i}_{=: \mu_s^T, \mathcal{F}_T\text{-measurable}} + \underbrace{\sum_{i=1}^n c_i \int_T^s e^{-x_i(s-u)} dW_u}_{=: Z_s, \text{ independent of } \mathcal{F}_T} \quad (53)$$

The future component Z_s is Gaussian with mean zero and variance

$$\sigma_{\text{cond}}^2(s, T) := \text{Var}(Y_s \mid \mathcal{F}_T) = \text{Var}(Z_s) = \sum_{i=1}^n \sum_{j=1}^n c_i c_j \frac{1 - e^{-(x_i + x_j)(s-T)}}{x_i + x_j} \quad (54)$$

Since $Y_s = \mu_s^T + Z_s$ with $Z_s \sim N(0, \sigma_{\text{cond}}^2)$ independent of $\mathcal{F}_T$, the moment generating function of the Gaussian gives

$$\mathbb{E}[e^{a Z_s}] = e^{\frac{1}{2} a^2 \sigma_{\text{cond}}^2(s, T)} \quad (55)$$

and therefore

$$\begin{aligned} \mathbb{E}[V_s | \mathcal{F}_T] &= \xi_0(s) \exp\left(a \mu_s^T - \frac{1}{2} a^2 \text{Var}(Y_s)\right) \mathbb{E}[e^{a Z_s}] \\ &= \xi_0(s) \exp\left(a \mu_s^T + \frac{1}{2} a^2 \sigma_{\text{cond}}^2(s, T) - \frac{1}{2} a^2 \text{Var}(Y_s)\right) \end{aligned} \quad (56)$$

where $\text{Var}(Y_s) = \sigma_{\text{total}}^2(s)$ is the total (unconditional) variance given by (36) evaluated at time s .

Conditional expectation in the rBergomi model. An analogous decomposition holds. The Volterra process $\widetilde{W}_s = \int_0^s (s-u)^{H-1/2} dW_u$ is split as

$$\widetilde{W}_s = \underbrace{\int_0^T (s-u)^{H-1/2} dW_u}_{=: \mu_s^T, \mathcal{F}_T\text{-measurable}} + \underbrace{\int_T^s (s-u)^{H-1/2} dW_u}_{=: Z_s, \text{ independent of } \mathcal{F}_T} \quad (57)$$

with $\text{Var}(Z_s) = (s-T)^{2H}/(2H)$. The conditional expectation follows the same Gaussian moment generating function argument, yielding

$$\mathbb{E}[V_s | \mathcal{F}_T] = \xi_0(s) \exp\left(a \mu_s^T - \frac{1}{2} \eta^2 [s^{2H} - (s-T)^{2H}]\right) \quad (58)$$

where the correction term $s^{2H} - (s-T)^{2H}$ equals $\text{Var}(\widetilde{W}_s) - \text{Var}(Z_s)$. The conditional mean μ_s^T is computed as a matrix-vector product between the stored increments $\Delta W_0, \dots, \Delta W_{k_T-1}$ and the kernel weights $(s-t_j)^{H-1/2}$.

Numerical integration. In both models, the integral in (52) is approximated by the trapezoidal rule on $M+1$ equally spaced points $s_0 = T, s_1, \dots, s_M = T + \Delta$:

$$\frac{1}{\Delta} \int_T^{T+\Delta} \mathbb{E}[V_s | \mathcal{F}_T] ds \approx \frac{1}{M} \sum_{m=0}^M w_m \mathbb{E}[V_{s_m} | \mathcal{F}_T] \quad (59)$$

where $w_0 = w_M = \frac{1}{2}$ and $w_m = 1$ for $1 \leq m \leq M-1$. We use $M = 15$ quadrature points. One VIX sample per Monte Carlo path is then

$$\boxed{\text{VIX}_T^{(m)} = 100 \sqrt{\max\left(\frac{1}{M} \sum_{m'} w_{m'} \mathbb{E}[V_{s_{m'}} | \mathcal{F}_T^{(m)}], 0\right)}} \quad (60)$$

3.2.7 VIX option pricing

Additive shift to match the market future. The VIX is not a directly tradable asset; VIX options are priced and quoted relative to the corresponding VIX futures contract. The market-observed VIX future for maturity T is denoted F_T^{mkt} . In general, the Monte Carlo average does not exactly coincide with F_T^{mkt} , due to model misspecification and finite-sample effects. To ensure consistency with the observed futures level, we apply an additive shift:

$$\text{VIX}_T^{(m), \text{shifted}} = \text{VIX}_T^{(m)} + (F_T^{\text{mkt}} - \bar{\text{VIX}}_T) \quad (61)$$

After shifting, the sample mean equals F_T^{mkt} exactly, so the model is used for the shape of the VIX distribution (and hence the smile) while the level is anchored to the market future.

Call price estimation. The price of a European call on VIX with strike K and maturity T is estimated as

$$\hat{C}_{\text{VIX}}(K, T) = e^{-rT} \frac{1}{N_{\text{paths}}} \sum_{m=1}^{N_{\text{paths}}} \max(\text{VIX}_T^{(m), \text{shifted}} - K, 0) \quad (62)$$

3.2.8 Implied volatility inversion

The Monte Carlo call prices obtained in Sections 3.2.5 and 3.2.7 are converted into implied volatilities by numerically inverting the appropriate closed-form pricing formula.

Black-Scholes inversion for SPX options. The Black-Scholes call price as a function of volatility is

$$C_{\text{BS}}(\sigma) = S_0 \Phi(d_1) - K e^{-rT} \Phi(d_2) \quad (63)$$

where

$$d_1 = \frac{\ln(S_0/K) + (r + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}, \quad d_2 = d_1 - \sigma\sqrt{T} \quad (64)$$

and Φ denotes the standard normal CDF. The implied volatility σ^{imp} is the unique positive root of

$$C_{\text{BS}}(\sigma^{\text{imp}}) = \hat{C}(K, T) \quad (65)$$

where $\hat{C}(K, T)$ is the Monte Carlo price from (51). This scalar equation is solved using Brent's method on the interval $[10^{-4}, 10]$.

Black-76 inversion for VIX options. Since VIX options are quoted on a futures contract, the Black-76 model is used in place of Black-Scholes. The Black-76 call price as a function of volatility is

$$C_{76}(\sigma) = e^{-rT} \left[F \Phi(d_1) - K \Phi(d_2) \right] \quad (66)$$

where $F = F_T^{\text{mkt}}$ is the market VIX future and

$$d_1 = \frac{\ln(F/K) + \frac{1}{2}\sigma^2 T}{\sigma\sqrt{T}}, \quad d_2 = d_1 - \sigma\sqrt{T} \quad (67)$$

The role of F in Black-76 is analogous to that of S_0 in Black-Scholes. The implied volatility σ^{imp} is obtained by solving

$$C_{76}(\sigma^{\text{imp}}) = \hat{C}_{\text{VIX}}(K, T) \quad (68)$$

3.3 Calibration

For the calibration we use the Differential Evolution algorithm from `scipy.optimize`, a global, derivative-free optimizer that maintains a population of candidate solutions and evolves them through mutation and recombination; this choice is well suited to our setting because the objective function is non-differentiable (being computed via Monte Carlo), potentially multimodal, and defined over a low-dimensional parameter space.

3.3.1 SPX-only calibration

The SPX loss is the mean squared error (MSE) computed as a flat average over all calibration pairs, since there is an explicit constraint to select 20 strikes for each maturity within a range that includes some at-the-money options and some moderately out-of-the-money ones, as those that are too far OTM tend to be excessively noisy:

$$\mathcal{L}_{\text{SPX}}(\boldsymbol{\theta}) = \frac{1}{|S|} \sum_{(K,T) \in S} \left(\sigma_{\text{SPX}}^{\text{mod}}(K, T; \boldsymbol{\theta}) - \sigma_{\text{SPX}}^{\text{mkt}}(K, T) \right)^2 \quad (69)$$

where $\boldsymbol{\theta} = (H, \eta, \rho)$, $\sigma_{\text{SPX}}^{\text{mod}}(K, T; \boldsymbol{\theta})$ is the model implied volatility computed via the Monte Carlo and inversion procedure described in the previous section, and $\sigma_{\text{SPX}}^{\text{mkt}}(K, T)$ is the market implied volatility from the data. Define Θ the feasible set, the SPX-only calibration solves

$$\boldsymbol{\theta}_{\text{SPX}}^* = \arg \min_{\boldsymbol{\theta} \in \Theta} \mathcal{L}_{\text{SPX}}(\boldsymbol{\theta}) \quad (70)$$

Results. Leaving all three parameters free, the algorithm selected suboptimal solutions (we also tried the Nelder-Mead algorithm, but it exhibited the same behavior). Therefore, inspired by [2], we adopted an approach based on guessing the parameters. First, H was fixed at the value estimated from the ATM skews, namely 0.061. Then, knowing that ρ governs the movement of the minimum of the smile, it was set to -0.9 by trial and error after testing several values. The minimization was thus performed only with respect to η , and by subsequently trying different values, we confirmed that in this case the algorithm was indeed able to find the global minimum.

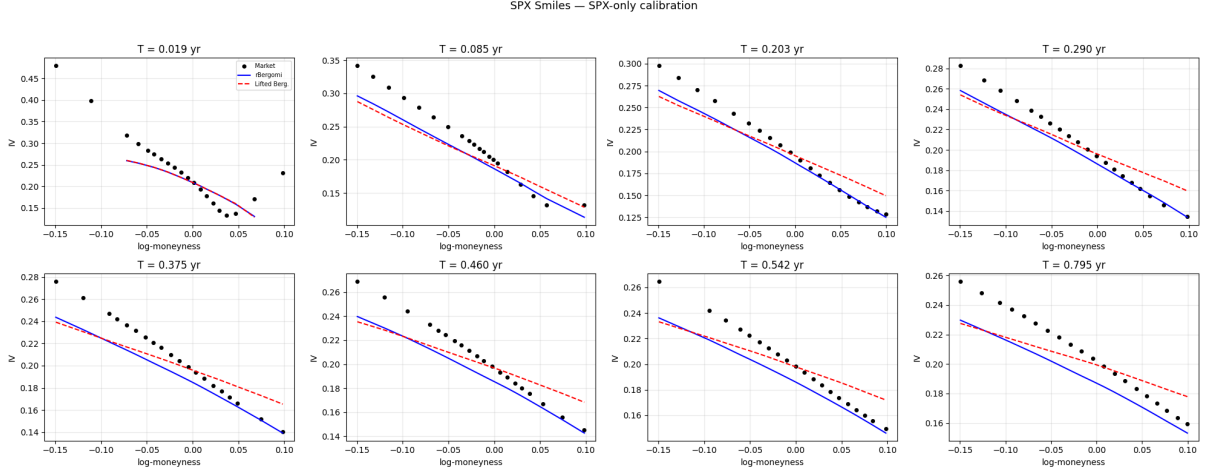


Figure 4: SPX smiles after SPX-only calibration

T	$\text{MSE}_{\text{rBerg}}$	$\text{MSE}_{\text{LiftBerg}}$
0.0192	0.000878	0.000886
0.0849	0.000594	0.000772
0.2027	0.000179	0.000355
0.2904	0.000163	0.000266
0.3753	0.000300	0.000313
0.4603	0.000241	0.000241
0.5425	0.000206	0.000217
0.7945	0.000287	0.000222
AVG	0.000356	0.000409

Table 1: Comparison of MSE for rough and lifted models calibrated on SPX

The calibrated models yield $\eta_{\text{rough}} = 1.863$ and $\eta_{\text{lifted}} = 1.661$, hence relatively close values. By performing a simulation with a high number of paths and time steps using these models, we

obtain the graphical results shown in Figure 4. First, we observe that for very short maturities (one week), the numerical implementation fails to reproduce the entire smile, likely due to issues in the inversion procedure, which returns NaN values. For the remaining maturities, as also shown in Table 1, we can see that the rough model achieves lower MSE values for intermediate maturities, while the lifted model performs better for longer maturities. We do not report the results for VIX smiles due to their poor performance, as the calibration is exclusively focused on fitting the SPX smiles.

3.3.2 VIX-only calibration

The equations are the same as before, but with VIX instead of SPX.

Results. In this case, the calibration yields $\eta_{\text{rough}} = 2.8681$ and $\eta_{\text{lifted}} = 3.5686$, which are noticeably higher than those obtained under the SPX-only calibration. From the graphical results, we can observe the effect of this change: there is a slight degradation in performance on the SPX smiles (especially for the rough model), in particular with an excessive downward shift that becomes more pronounced as maturity increases.

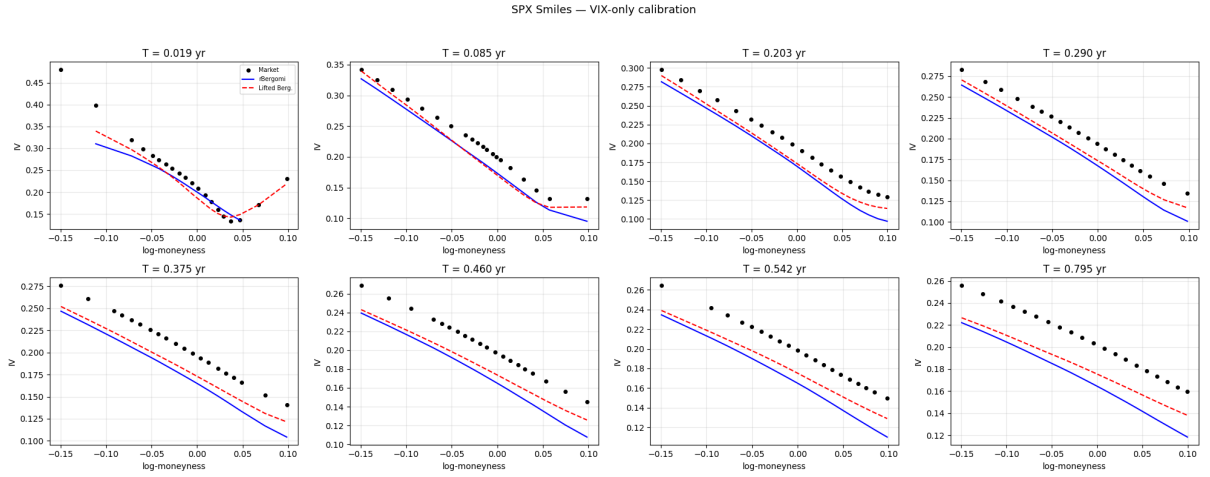


Figure 5: SPX smiles after VIX-only calibration

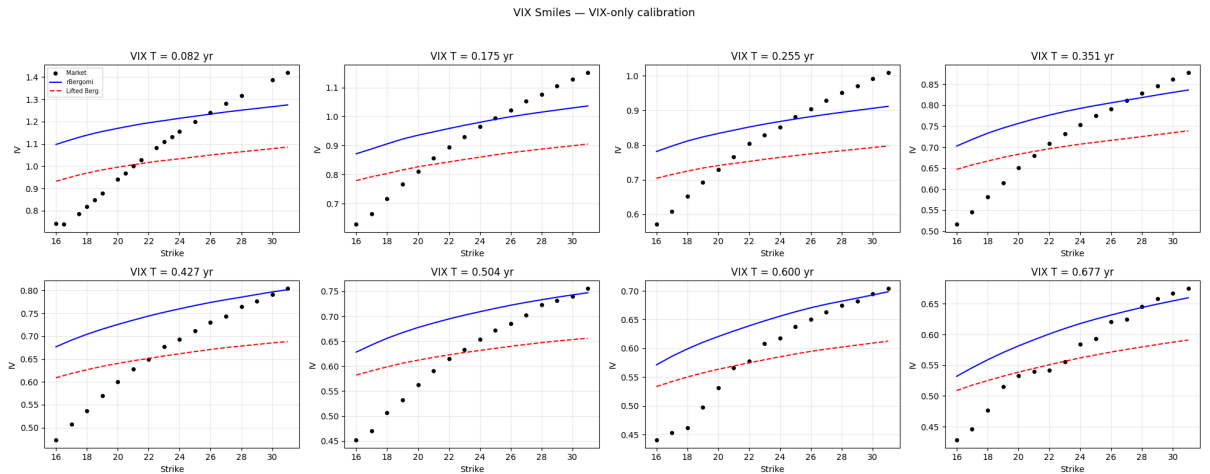


Figure 6: VIX smiles after VIX-only calibration

However, there is a trade-off, as performance improves for VIX smiles. The higher value of η

allows for a better adjustment of both the level and the steepness compared to the previous case. Nevertheless, the MSE values remain significantly higher than those for SPX smiles, highlighting the greater difficulty of Bergomi-type models in fitting VIX smiles. Moreover, unlike in the SPX case, the two models now produce noticeably different levels for the volatility curves, with the lifted model systematically underestimating volatility compared to the rough one.

T	$\text{MSE}_{\text{rBerg}}$	$\text{MSE}_{\text{LiftBerg}}$	T	$\text{MSE}_{\text{rBerg}}$	$\text{MSE}_{\text{LiftBerg}}$
0.0192	0.000754	0.000481	0.0822	0.043806	0.029032
0.0849	0.000540	0.000491	0.1753	0.014564	0.020481
0.2027	0.000642	0.000336	0.2548	0.010606	0.015628
0.2904	0.000668	0.000343	0.3507	0.008524	0.007658
0.3753	0.000992	0.000531	0.4274	0.011089	0.006142
0.4603	0.001024	0.000546	0.5041	0.008894	0.005398
0.5425	0.001161	0.000529	0.6000	0.005392	0.004176
0.7945	0.001394	0.000692	0.6767	0.002810	0.002739
AVG	0.000897	0.000494	AVG	0.013211	0.011407

Table 2: MSE comparison for SPX (left) and VIX (right) smiles under VIX-only calibration

3.3.3 Joint calibration

The joint calibration minimizes a weighted sum of the SPX and VIX losses. With $w_{\text{VIX}} = 1$ (equal weighting), yielding

$$\theta_{\text{joint}}^* = \arg \min_{\theta \in \Theta} \mathcal{L}_{\text{SPX}}(\theta) + w_{\text{VIX}} \mathcal{L}_{\text{VIX}}(\theta) \quad (71)$$

Results. We now obtain $\eta_{\text{rough}} = 2.8512$ and $\eta_{\text{lifted}} = 3.5431$, which are slightly lower than before. This highlights the compromise made to achieve a somewhat better fit for SPX, although the values remain relatively high. This indicates that the optimizer places significant emphasis on fitting the VIX, as it is more challenging than SPX. We do not report the results, as they are very similar to those obtained under the VIX-only calibration.

3.3.4 Final results

In Table 3, we report all calibration and performance results. Overall, we observe that the lifted model achieves more stable MSE values across different calibration settings and generally lower errors compared to the rough model. We can also note that, on average, the best performance is achieved through joint calibration.

	H	η	ρ	MSE_{SPX}	MSE_{VIX}	Time (s)
rBerg [SPX]	0.0610	1.8683	-0.9000	0.000356	0.035138	40.38
rBerg [VIX]	0.0610	2.8681	-0.9000	0.000897	0.013211	34.79
rBerg [Joint]	0.0610	2.8512	-0.9000	0.000881	0.012898	71.69
LiftBerg [SPX]	0.0610	1.6610	-0.9000	0.000409	0.168966	15.13
LiftBerg [VIX]	0.0610	3.5686	-0.9000	0.000494	0.011407	23.54
LiftBerg [Joint]	0.0610	3.5431	-0.9000	0.000482	0.011753	51.65

Table 3: Calibration results: parameters, MSE on SPX and VIX smiles, and calibration time

Regarding calibration time, it was measured using 500 Monte Carlo paths and 100 time steps, on an NVIDIA Tesla T4 GPU. We observe that lifted models train faster than rough ones,

confirming that the recovered Markovian structure of the model provides improved computational performance.

3.4 Regularity study

Finally, we verify that the simulated variance paths exhibit the expected rough behavior with the joint calibrated models. The procedure estimates the Hurst exponent H from the scaling properties of the increments of the log-variance process $\log V_t$. The key observation is that, for a process with fractional Brownian motion-like regularity, the q -th absolute moment of the increments scales as a power law in the lag:

$$m(q, \Delta) := \mathbb{E}[|\log V_{t+\Delta} - \log V_t|^q] \sim K_q \Delta^{\zeta_q} \quad (72)$$

with $\zeta_q = qH$ because the process is monofractal (i.e. governed by a single Hurst exponent). The estimation proceeds in three steps:

1. **Empirical moments.** Given N_{paths} simulated log-variance paths on a uniform grid with step Δt , we compute the empirical moment for each moment order $q \in \{0.5, 1, 1.5, 2, 3\}$ and each lag $\Delta = 1, 2, \dots, 100$ (in units of Δt):

$$\hat{m}(q, \Delta) = \frac{1}{N_{\text{paths}} (N_{\text{steps}} - \Delta)} \sum_{m=1}^{N_{\text{paths}}} \sum_{k=1}^{N_{\text{steps}} - \Delta} |\log V_{t_{k+\Delta}}^{(m)} - \log V_{t_k}^{(m)}|^q \quad (73)$$

2. **Scaling exponents.** For each q , we perform a linear regression of $\log \hat{m}(q, \Delta)$ against $\log \Delta$:

$$\log \hat{m}(q, \Delta) \approx \zeta_q \log \Delta + C_q \quad (74)$$

The slope ζ_q is the scaling exponent associated with moment order q .

3. **Hurst exponent.** Under the monofractality hypothesis, we estimate H as the slope of a linear regression of ζ_q against q .

Results. Figure 7 graphically illustrates this procedure for the lifted model. From the regression, we obtain $H_{\text{rough}} = 0.203$ and $H_{\text{lifted}} = 0.142$, confirming the roughness of volatility in our models. These values are somewhat higher than the one estimated from ATM skews, but this is not unexpected, as all such estimates rely on approximate methods.

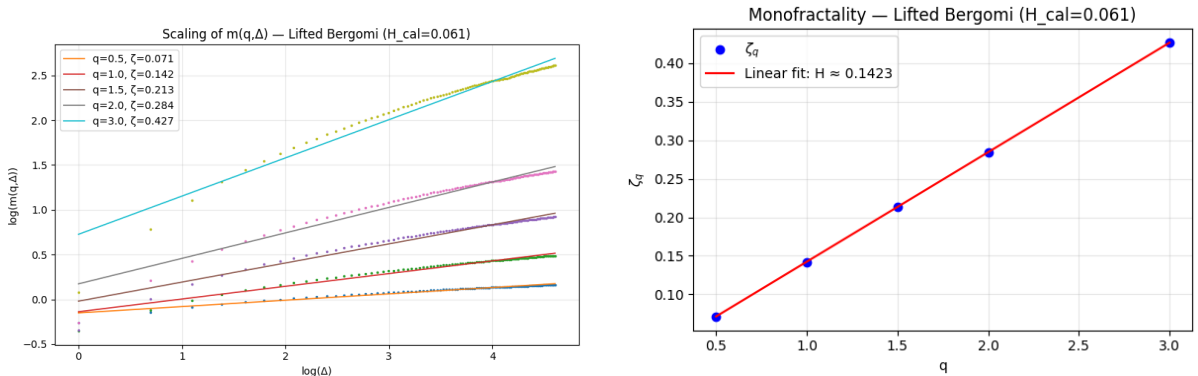


Figure 7: Scaling exponents and Hurst parameter via linear regression

In Figure 8, we also display a sample of variance trajectories generated by the models. We can indeed observe that these processes exhibit a rougher, more irregular dynamics compared to typical diffusion-type processes.

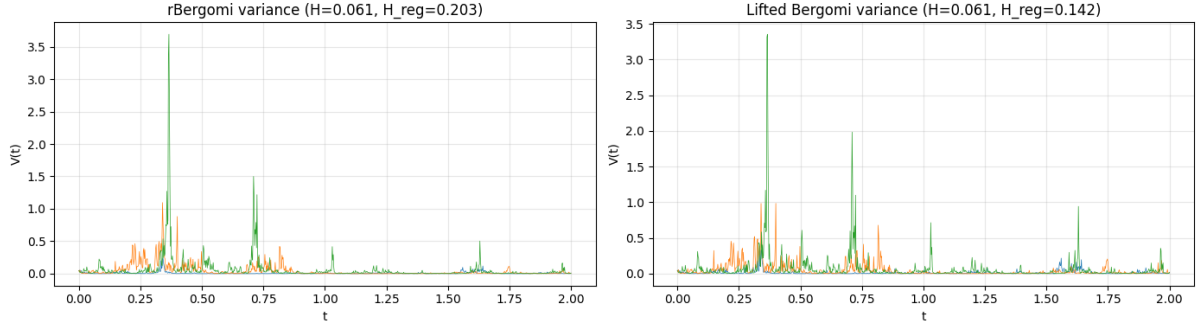


Figure 8: Variance trajectories sample

4 Conclusion and potential extensions

In conclusion, we have numerically implemented Bergomi-type models and, through the calibration results, we have been able to analyze the strengths and weaknesses of both the models themselves and our procedure, as well as the role played by the various parameters.

The imperfect fits, particularly for VIX smiles, may be due not only to the reasons discussed previously, but also to the fact that the data were collected during a period of market stress, driven by geopolitical and economic instability caused by the war in Iran. In particular, models without jumps, such as those considered here, struggle to adapt to abnormal market conditions where the underlying model assumptions are no longer consistent with the real phenomena we aim to model.

Possible extensions include, for instance, the use of variance reduction techniques such as control variates or antithetic variables in Monte Carlo simulations, as well as considering alternative models such as the Heston model, local volatility models or jump models to provide a more comprehensive comparison.

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